

Luka Kawashima

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EDUCATION

California State University, East Bay

Bachelor of Science in Computer Science

Hayward, CA

Expected Fall 2027

EXPERIENCE

Undergraduate Research Assistant

Aug. 2026 - Present

California State University, East Bay (Advisor: Dr. Fay Zhong)

Hayward, CA

- Developed an initial VR memory-task prototype using Unity, C#, and XR Interaction Toolkit, with timed word presentation, delayed recall prompts, and response logging.
- Reviewed and summarized literature on VR-based programs and game design features intended to support cognitive stimulation in older adults, informing early-stage project planning with a faculty advisor.

PROJECTS

Arkvain on Steam – Multiplayer FPS | *C#, Unity, PurrNet, Steamworks API*

- Built a multiplayer FPS with full client-side prediction and server rollback in C# using Unity and the PurrNet networking library; attracted 200+ wishlists.
- Integrated Steam P2P lobby hosting, server browsing and join codes with synchronized per-player loadouts, cosmetics, map, and game-mode selection.
- Implemented server-authoritative free-for-all and rotating 1v1 modes, including spawning, respawning, player queues, spectator support, and disconnect handling.

play4dchess.vercel.app – Browser Game and Engine | *TypeScript, JavaScript, WebGL, three.js, Web Workers*

- Built a chess program that renders 4D chessboards through custom 4D-to-3D projections with instanced three.js geometry.
- Implemented a time-bounded computer opponent using iterative-deepening search, alpha-beta pruning, and quiescence search, offloading computation to a Web Worker to preserve UI responsiveness.
- Configured GitHub Actions CI for automated testing and production builds, with continuous deployment through Vercel.

Umbra Tags - Desktop Image File Manager | *C#, .NET, WinForms*

- Built a Windows desktop application and put into active personal use for over a year and running to organize image downloads across several local collections with hierarchical tags.
- Implemented perceptual-hash duplicate guarding, autosaving, and temporary-file backups to protect library metadata.

TECHNICAL SKILLS

Languages: C#, JavaScript, TypeScript, C++, Python, GLSL/HLSL, HTML/CSS

Frameworks & Libraries: Unity, .NET, WinForms, three.js, PyTorch, NumPy, PurrNet/PurrDiction, XR Interaction Toolkit

Graphics & APIs: WebGL, OpenGL, Steamworks API

Tools & Platforms: Git, GitHub, Vercel, Cloudflare Pages